

Markscheme

Mathematics: analysis and approaches

Practice question bank

58. (a)

Consider the function $f(x) = 3^x + 3^{-x}$. Suppose $f(x) = 3.5$ for some value(s) of x .

Letting $u = 3^x$, form a quadratic equation in u and hence find the two possible values of u .

$$3^x + 3^{-x} = u + \frac{1}{u} = 3.5 \quad (M1)$$

$$u^2 - 3.5u + 1 = 0 \quad \text{A1}$$

$$\text{using the quadratic formula: } u = \frac{3.5 \pm \sqrt{8.25}}{2} \quad (M1)$$

$$u = 3.19 \text{ or } u = 0.314 \quad \text{A1}$$

[9 marks]

(b)

Hence find the two corresponding values of x , correct to 3 significant figures.

$$x = \log_3(3.186 \dots) \quad (M1)$$

$$x_1 = 1.05 \quad \text{A1}$$

$$x = \log_3(0.3139 \dots) \quad (M1)$$

$$x_2 = -1.05 \quad \text{A1}$$

[4 marks]

(c)

Hence state the value of $x_1 + x_2$.

$$\text{since } u_2 = \frac{1}{u_1}, \text{ the two } x \text{ values are equal in magnitude and opposite in sign, so } x_1 + x_2 = 0 \quad \text{R1}$$

[1 mark]